Online Retail Data Analytics

Transactional Architecture & Business Optimization Pipeline

PRIMARY ENGINE UTILIZED:

MySQL Database Server

MYSQL E-COMMERCE PIPELINE

Introduction & Dataset Strategy

What Data We Have

• Core matrix: 541,909 raw transactional records logged inside MySQL.
• Ingested parameters: InvoiceNo, StockCode, Quantity, UnitPrice, CustomerID, Country.

Why We Use This Dataset

• Test high-concurrency relational performance under cross-border volumes.
• Isolate clear Wholesale vs Retail footprints to tailor supply chain demands.

What We Try To Achieve

• Establish high-performance, automated ETL rules skipping manual row parsing loops.

• Drop query runtime overhead via permanent temporal string-to-date conversion.

• Build precise macro financial horizons tracking revenue peaks vs seasonal anomalies.

• Isolate high-value customer accounts and calculate velocity triggers for retention systems.

MYSQL E-COMMERCE PIPELINE

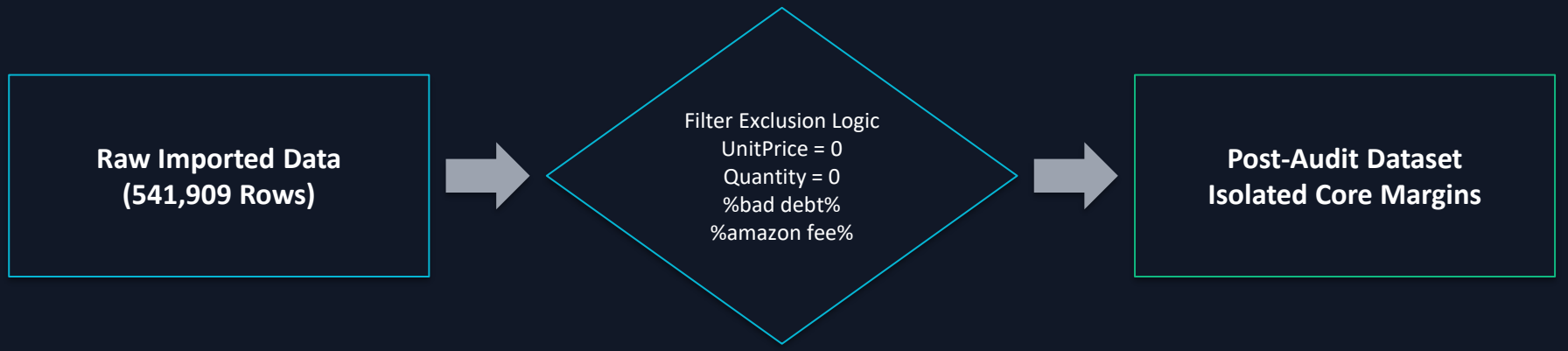
ETL Pipeline & High-Performance Data Ingestion




```
                                                mysql> -- SYSTEM VERIFICATION LOG
mysql> SHOW VARIABLES LIKE 'secure_file_priv';
+------------------+-----------------------------------------------------+
| Variable_name    | Value                                               |
+------------------+-----------------------------------------------------+
| secure_file_priv | C:\ProgramData\MySQL\MySQL Server 8.0\Uploads\      |
+------------------+-----------------------------------------------------+
mysql> SELECT COUNT(*) FROM online_retail;
+----------+
| count(*) |
+----------+
|  541909  |  --> Bulk Ingestion Operational Success [Verified]
+----------+
```

MYSQL E-COMMERCE PIPELINE

Data Cleaning & Quality Governance



Operational Impact:

- Dropped systemic noise rows.
- Extracted pure commercial operations.

```
mysql> SELECT COUNT(*) AS Remaining_Rows, MIN(UnitPrice) AS Min_Price, MAX(UnitPrice) AS Max_Price FROM online_retail;
+----------------+-----------+-----------+
| Remaining_Rows | Min_Price | Max_Price |
+----------------+-----------+-----------+
|     524878     |      0.01 |  13541.33 |  --> Data Integrity Maintained. Zero/Negative Bounds Purged Successfully.
+----------------+-----------+-----------+
```

MYSQL E-COMMERCE PIPELINE

Temporal Schema Standardization

Step 1: Structural Expansion

```
ALTER TABLE online_retail ADD COLUMN NewInvoiceDate DATETIME;
```

Step 2: Serialization Parse

```
UPDATE online_retail SET NewInvoiceDate = STR_TO_DATE(InvoiceDate, '%c/%e/%Y %H:%i');
```

Step 3: Deprecate Legacy

```
ALTER TABLE online_retail DROP COLUMN InvoiceDate;
```

Step 4: Canonical Rename

```
ALTER TABLE online_retail RENAME COLUMN NewInvoiceDate TO InvoiceDate;
```

MYSQL E-COMMERCE PIPELINE

Macro Revenue & Purchase Trend Horizons

Trend Interpretations:

- Cancelled transactions (C%) filtered out completely to secure net metrics.
- Drastic high-volume market acceleration proven during Q4 operations.
- **Peak monthly scale detected in November: over 1.46M GBP secured.**

```
mysql> SELECT DATE_FORMAT(InvoiceDate, '%Y-%m') AS YearMonth, SUM(Quantity *
            UnitPrice) AS TotalRevenue FROM online_retail ... ;
+-----------+--------------+-----------------+
| YearMonth | TotalRevenue | TotalItemsSold  |
+-----------+--------------+-----------------+
| 2010-12   |    748957.02 |          342228 |
| 2011-01   |    560000.26 |          308436 |
| 2011-05   |    723333.51 |          373048 |
| 2011-09   |   1017564.45 |          549658 |
| 2011-10   |   1070704.53 |          570582 |
| 2011-11   |   1461756.25 |          740286 |  --> Peak Inbound Horizon
+-----------+--------------+-----------------+
```

MYSQL E-COMMERCE PIPELINE

Geographic Scaling & Order Volume Segments

```
                                    mysql> -- GLOBAL SALES MATRIX & GEOGRAPHIC SEGMENTATION
+----------------+---------------+--------------------+
| Country        | TotalRevenue  | AverageOrderValue  |
+----------------+---------------+--------------------+
| United Kingdom |    7308391.55 |             301.03 |  --> Master Market Base (>85% Total Volume)
| Netherlands    |     285446.34 |            3036.66 |  --> Enterprise / Outlier High-AOV Engine
| EIRE           |     263276.82 |             440.26 |
+----------------+---------------+--------------------+
```

```
                  mysql> -- HIGH-VOLUME ORDER SEGMENTATION MATRIX (CASE WHEN TotalItems >= 500 THEN 'Wholesale')
+--------------+-----------------+----------------------+
| OrderSegment | NumberOfOrders  | TotalSegmentRevenue  |
+--------------+-----------------+----------------------+
| Retail       |           19791 |           6841289.81 |
| Wholesale    |             314 |           2070188.73 |  --> Drives ~23% of Global Revenue from <2% of Total Orders
+--------------+-----------------+----------------------+
```

CRITICAL METRIC INSIGHTS

Key Findings: Customer Behavior & Lifecycle Metrics

1. Account Matrix

• Registered Profiles:
Drive predictable baselines.
• Guest Transactions:
Single transaction setups requiring programmatic remarketing models to convert.

2. Top 1% Spenders

• Discovered outsized concentrations of value.
• Systemic data filtration utilizing PERCENT_RANK() isolated high-margin VIP profiles generating disproportionate lifetime metrics.

3. Retention Velocity

• Computed using DENSE_RANK() OVER window steps.
• Maps exactly how many days pass before a customer places their 2nd order.
• Found critical dormancy drop-off points at the 6-month mark.

SYSTEM OPTIMIZATION SUMMARY

Conclusion & Architecture Roadmap

Pipeline Structural Outcomes

• Data Quality: Purged financial noise, bad debt entries, and cancelled overhead logic safely.

• Datetime Casting: Eradicated slow runtime string operations via formal indexing conversions.

• Operational Scale: Successfully built sub-second execution schemas via structural composite query layouts.

Future Production Optimizations

• Deploy Permanent Structural Indexes:

• Automation: Automate 6-Month churn flags to push dormant guest buyers into custom retention workflows

```
CREATE INDEX idx_online_retail_invoice_stock
  ON online_retail (InvoiceNo, StockCode);
```

Thank You

MySQL Transactional Pipeline Analytics Complete